Thesis erratum

Thesis. Minjun Kim, *Electrical measurements and prediction of dielectric barrier surface damage in nonthermal atmospheric pressure dielectric barrier discharge plasma based on machine learning*, M.S. thesis, Kwangwoon University, 2023.

Date of this erratum. 2026-11-01.

Basis.

- Each item refers to an entry of the verification report, given by its ID (A1, C5, …).
- "Re-analysis" values come from device m2 (40 normal and 120 degraded captures).
- Items 59–65, and the revised items 2, 4 and 47, are based on the complete raw-data archive with the undamaged device M3 as a control (raw-data verification; review section J).

Page numbers. These are the numbers printed on the thesis pages. The thesis is written in Korean, so the original wording is quoted in Korean only in the Korean version.

Types of item.

Type	Meaning
Correction	Error of fact, arithmetic or definition
Addition	Missing interpretation or limitation
Notation	Units, typos, formatting

> The thesis PDF itself is unchanged. Read it together with this erratum and the verification report.

Abstract (pp. v–vi)

#	Location	Correction	Type · ref
1, 3	KO and EN abstract	Replace "discharge energy per second" with "**discharge energy per cycle**".	Correction · C1
2, 4	KO and EN abstract, sentence on changes due to damage	Replace with: "Surface damage lowered the effective voltage. Compared with an undamaged device measured in the same sessions, the damage-specific decrease was about 3 % (≈ 48 V); about half of the raw decrease was a supply variation common to all devices. The capacitances and the discharge charge tended to increase, but not significantly against the control. The effective current, the number of discharges and the electron density depended on the oscilloscope settings and showed no significant damage-specific change."	Correction · D1, J

Chapter 1

#	Page	Correction	Type · ref
5	1	Write "electron temperature of a few eV (≈ 10^4–10^5 K), gas temperature ≈ 300 K" instead of "> 2000 °K / 300 °K".	Notation · A5
6	1	Add: "electron density in atmospheric DBD micro-discharge channels is 10^{14}–10^{15} cm^{-3} [61]". The quoted 10^{10}–10^{13} cm^{-3} applies to low-pressure plasmas.	Addition · A5
7	3	Eq. (7) is **Townsend's semi-empirical formula**, not the Saha ionisation formula.	Correction · A1
8	5	Below the Paschen curve there is no breakdown. To the right of the minimum, collisions are too frequent for electrons to gain the ionisation energy; to the left, there are too few collisions.	Correction · A4
9	6	For a 100 µm air gap, Eq. (8) gives a breakdown voltage of ≈ 0.7–0.9 kV, not "several kV". The applied voltage must reach ≈ 2 kV peak because of the capacitive divider formed by the dielectric.	Correction · A3

Chapter 2

#	Page	Correction	Type · ref
10–11	11	The HV probe is the **P6015A**; the current probe is the **P6021A**.	Correction · B1
12	11–12	Add the measurement limits: sampling 10 ns (some captures 20 ns), 5000 points (≈ 2.3 periods), 8 bit (40 V/LSB; 1.6–8 mA/LSB), current-probe bandwidth 60 MHz. As a result, pulse counts and peak currents are lower bounds.	Addition · B2
59	12	Add to Sec. 2.1.3: a third device (M3) was operated in parallel for 60 h and measured in the same sessions, without visible damage. It serves as a control for supply and environmental variations.	Addition · J
60	11–12	Add: the current scale (0.04–0.4 A/div) and the sampling interval (10 / 20 ns) changed between and within sessions. Within a session, I_rms rises by 3.5 mA per doubling of A/div, and 20 ns sampling gives 6.6 fewer pulses. Compare I_rms, peak current, number of discharges and n_e only at fixed settings.	Addition · J
61	13	Fig. 7, "the discharge current rose from about 100 mA to 250 mA": state that Fig. 7 is a single example. At the same current scale (0.1 A/div) the peak currents are 251 → 272 mA (M2, not significant) and 252 → 213 mA (M1).	Correction · J

Chapter 3

#	Page	Correction	Type · ref
13	16	V_min is ≈ 300–700 V (mean 504.5 ± 168.8 V), not "300–500 V".	Notation · C3
14, 16	16–17	"Energy per second" → "**energy per cycle**": 0.13 ± 0.022 mJ (normal) and 0.13 ± 0.024 mJ (degraded). Add the mean power P = E·f ≈ **6.0 W**.	Correction · C1

#	Page	Correction	Type · ref
62	16	V_rms "1.5 kV → 1.4 kV": add that the undamaged device M3 dropped in the same sessions (e.g. 1502 → 1369 V from 10 to 15 h). The damage-specific decrease is ≈ 48 V.	Addition · J
15, 34	16, 32–33	Write mean ± SD in a single unit: 1.50 ± 0.017 kV, 20.8 ± 4.0 mA, 34.2 ± 4.9 pF, 11.3 ± 1.2 pF, 17.3 ± 2.8 pF, 59.4 ± 9.0 nC, etc.	Notation · C10
17	17	Eq. (11) → $\mathbf{E = \int_0^T v\, i\, dt}$ [J/cycle] and P = E/T [W].	Correction · C1
18–20	19–21	Replace Eqs. (13)–(15) with **text A** below. 55.6 nC / 30.5 pF = 1.82 kV, not 1.5 kV. The original Eq. (15) evaluates to V_max (2.1/2.0 kV). The burning voltage is U_b = 860 ± 68 V (normal) and 811 ± 75 V (degraded).	Correction · C4, C5
21	22	Describe the pulse detector exactly: threshold = median prominence × 1.5, and pulses with IQR-outlier widths removed. Note that this threshold is relative to each capture.	Addition · C11
22	23	Fig. 14 shows T_3–T_4, not T_1–T_2.	Notation · C3
23, 25	24–25	In Eqs. (18) and (20), use **U_b** instead of V_breakdown. n_e then becomes ≈ 3.9×10^{12} (normal) and 4.0×10^{12} cm^{-3} (degraded), with no significant difference. Subtracting the displacement current from i_plasma is recommended.	Correction · C7
24	24	The area 4.5×10^{-4} cm² is the pad top area, used as an approximation of the current cross-section. n_e is therefore a cell-averaged relative index.	Addition · C7
26	26	Use N = p/(k_BT) = 2.50×10^{19} cm^{-3} (1 atm, 20 °C). This is the gas number density, not "Avogadro's number".	Correction · C6
27	26	E/N ≈ 34×10^{-16} V cm² (≈ 340 Td) with U_b. Note that even the original inputs give 75.0, not 81.38.	Correction · C6
28	27	Within the data range of Fig. 15, D/μ ≈ 6 V, so T_e ≈ **6 eV**. Re-analysis: 6.2 ± 0.3 eV (normal), 6.0 ± 0.4 eV (degraded). The original 13 eV was read from the extrapolated part of the curve.	Correction · C6
29	26	Add: D/μ is the characteristic energy; it equals k_BT_e/e only for a Maxwellian distribution.	Addition · C6
30	28	Eq. (23) → $\mathbf{Q(t) = \int_0^t i\, dt' + Q_0}$, after removing the current offset.	Correction · C2
31–33	32	Table 2: the C_total fitting error is 7.63×10^{-14} F. Write $p < 10^{-15}$ instead of 1.11×10^{-16}. Add $R^2 \approx 0.98$ and note the autocorrelation of the fitted points.	Correction · C8
35	32	Add: the discharge-phase slope is an effective ζ_diel ≤ C_d when the discharge covers only part of the dielectric [50].	Addition · C9

Text A (replaces Sec. 3.1.3, Eqs. 13–15)

- The gas-gap voltage is u_g(t) = v(t) − Q(t)/C_dielectric (13′). During discharge it stays approximately at the burning voltage U_b [42, 55].
- At ignition, the wall charge Q_on left over from the previous half cycle is still on the dielectric. It produces V_wall = −Q_on/C_dielectric (14′).
- Hence U_b = V_min − Q_on/C_dielectric = (1 − C_total/C_dielectric)·(V_max + V_min)/2 (15′).
- With the mean values of Table 2, U_b ≈ 0.92 kV, consistent with the Paschen value for a 100 μm air gap (0.72–0.87 kV). Re-analysis gives 860 ± 68 V (normal) and 811 ± 75 V (degraded).
- Why the original formula fails: using the transferred charge Q_max = C_dielectric (V_max − V_min) as the wall charge gives V_min + Q_max/C_dielectric = V_max.

Chapters 4–5

#	Page	Correction	Type · ref
36	40	"Degradaion" → "Degradation".	Notation
63	36, 46	Add to Sec. 4.2 and Fig. 27 / Table 5: the M1 micrographs taken after 5 h (before the 10 h measurement) already show a melted blister at an electrode end. The M1 5 h and 10 h data labelled normal may contain a damaged cell.	Addition · J
64	38–40	Figs. 23–24 "(a) normal discharge electrode" → "(a) reference sample (Ref)". Table 3: "[%]" → "**[at.%]**"; add that each column is a single 10,000× map, and give the range over all maps (damaged sites 1.3–13.9 at.% Ag, intact dielectric 0.0–0.7 at.%).	Correction · J
37	43	"Freqency" → "Frequency"; the unit of the electron-density SD is cm^{-3}.	Notation
38	43	Add: the features are not independent. C_gas follows from C_d and C_total; T_e and n_e follow from the breakdown voltage.	Addition · D9
39	44–45	Remove the references to a DNN (no results are reported).	Notation · D9
40	45	Add: the training/test set is a random split of time-ordered captures from one device, so test scores are optimistic. The cross-device validation set gives the primary performance.	Addition · D5
41	47	"min_samples_splite" → "min_samples_split".	Notation
42–44	49–50	Precision = fraction of samples predicted *degraded* that are degraded. Recall = fraction of truly degraded samples that are predicted degraded. TP, TN, FP, FN refer to the classes degraded (positive) and normal (negative); **FN = degraded predicted as normal**.	Correction · D3
45	51	Table 6: precision and recall are **macro averages**; F1 is the **binary F1** of the degraded class.	Correction · D2
46	51	Add: on the validation set, LR and SVM reach AUC 0.99 but only 76.4 % accuracy. This is a threshold shift between devices, which per-device baseline normalisation removes.	Addition · D4
47	52	Add: V_rms alone almost separates the classes (AUC ≈ 1.0); the classifier mainly detects the drop of the applied voltage. About half of that drop also occurs in the undamaged device M3 measured in the same sessions.	Addition · D6, J
65	51–52	Add: classifiers trained on M2 call 95–100 % of the undamaged M3's captures "degraded" in sessions with a low supply voltage. Report the false-positive rate on a control device together with the scores of Table 6.	Addition · J
48	53	A random forest **averages** the votes of all 1000 trees; it does not select the best tree. Fig. 29 shows one tree.	Correction · D7
49	55	Tables 7–8 show no drop of the "normal" probability before damage (0.886 / 0.909; 0.816 / 0.818). The "≥ 10 % drop" precursor criterion is therefore not supported and should be stated as future work.	Correction · D8
50	56	"Fig. 32 (a)/(b)" → "Fig. 30 (a)/(b)".	Notation

Chapter 6 (p. 57)

#	Correction	Type · ref
51–52	Same changes as items 1–2 (energy per cycle; the list of significant and non-significant changes).	Correction · D1, J

References

#	Correction
53	[25] duplicates [10] (Michel et al., J. Appl. Phys. 105, 073302 (2009)); merge them.
54–55	In [26] and [27], remove the affiliation letters from the author names: Gimeno, Hanson, Goldhawk, Simpson, Lavoie.
56	[28]: add the journal, *J. Appl. Phys.* 124, 183301 (2018).
57	[13]: add the title, journal, volume, pages and year.
58	[1]: check the edition (Plenum 2nd ed. 1984, or Springer 3rd ed. 2016).